M&Ms-2 Segmentation Report

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Abstract

Purpose: We aimed to develop and evaluate deep learning models for automated cardiac segmentation and pathology classification in cardiac MRI. This addresses the need for reliable, accurate tools that work across various heart conditions, following the M&Ms-2 challenge guidelines.

Methods: We built and compared three approaches. First, a temporal-aware 2D+Time UNet that uses 5-frame windows from CINE MRI sequences to better grasp heart motion. Second, a two-stage transformer pipeline: SegFormer-B0 handles 4-class segmentation on 256×256 slices, followed by ViT-B/16 for disease classification using the segmented areas. Third, standard 2D UNet models as baselines. For preprocessing, we aligned sequences for the temporal method and normalized intensities across all. We trained on the M&Ms-2 dataset with train-validation splits, using Dice plus CrossEntropy losses and AdamW optimization.

Results: Each method improved in different ways. The 2D+Time UNet led in segmentation by using motion context, leading to sharper boundaries and consistent anatomy through cardiac cycles. In contrast, the SegFormer-ViT pipeline excelled at handling multi-center data variations and delivered solid pathology classification from segmented features. Traditional 2D UNets offered reliable baselines. Overall, all tackled dataset diversity and pathology challenges effectively.

Conclusions: This study highlights flexible, deep learning options for cardiac MRI analysis. Temporal models stand out for motion handling, transformers for robust classification across pathologies, and our comparisons guide choices for real clinical needs. These advances surpass basic 2D limits and pave the way for standardized tools in varied heart conditions.

Keywords: Cardiac MRI, M&Ms-2 challenge, Deep learning, UNet, SegFormer, Vision Transformer, Temporal segmentation, CINE MRI, Multi-class segmentation, Pathology classification, Cardiac motion, Right ventricle, Medical image analysis

1 Introduction

Assessment of cardiac structures by MRI is a critical but challenging part of cardiovascular imaging workflows. Manual delineation of ventricular endocardial and epicardial borders for volumetric quantification is labour-intensive and exhibits significant variability. Cardiac segmentation presents unique challenges: complex geometries, variable wall thickness, temporal dynamics across cardiac cycles, and lower contrast-to-noise ratios in steady-state free precession (SSFP) cine sequences make automated boundary detection inherently difficult. In addition, pathological conditions such as pulmonary hypertension, arrhythmogenic right ventricular cardiomyopathy (ARVC), and congenital heart disease introduce significant morphological variations that traditional approaches struggle to accommodate.

Current clinical practice relies on time-intensive manual segmentation with substantial inter-reader variability, creating a clear need for robust, automated solutions that can maintain clinical-grade accuracy while reducing analysis time and improving measurement reproducibility across diverse pathological presentations. The temporal nature of cardiac MRI, particularly in CINE sequences, provides rich information about cardiac motion and function that has been underutilized in traditional frame-by-frame analysis approaches.

This study addresses these challenges through a comprehensive evaluation of multiple deep learning architectures, ranging from temporal-aware convolutional approaches to state-of-the-art transformer-based methods, providing insights into optimal architectural choices for different clinical scenarios and computational constraints.

2 Background

Cardiac MRI provides the reference standard for ventricular functional assessment, offering superior temporal resolution (25-50 phases per cardiac cycle) and spatial resolution (1.4-2.0mm in-plane) compared to echocardiography. The M&Ms-2 challenge dataset encompasses 360 subjects across multiple pathologies and imaging centers, providing ground truth annotations for ventricular boundaries across diverse clinical presentations including dilated cardiomyopathy, hypertrophic cardiomyopathy, ARVC, and pulmonary hypertension.

Traditional automated segmentation approaches for cardiac MRI have evolved through several paradigms. Early methods relied on statistical shape models and atlas-based approaches, which struggled with pathological variations. The introduction of deep learning, particularly U-Net architectures with encoder-decoder structures and skip connections, achieved reasonable performance on relatively homogeneous datasets but often struggled with domain shift in multi-center acquisitions and significant anatomical variations in pathological cases.

Recent advances have explored two key directions: (1) temporal modelling approaches that leverage the inherent temporal structure of CINE MRI sequences to improve segmentation consistency and accuracy, and (2) transformer-based architectures that capture long-range spatial dependencies through self-attention mechanisms. The temporal dimension of cardiac MRI has been particularly underexplored, despite containing rich information about cardiac motion patterns that could significantly enhance segmentation performance.

The SegFormer architecture introduces a paradigm shift by replacing convolutional encoders with hierarchical vision transformers. Its key innovation lies in an efficient MLP decoder that aggregates multi-scale features without requiring complex pyramid pooling modules. For medical imaging, this offers: (1) improved global context handling, (2) reduced sensitivity to local texture variations, and (3) more efficient multi-scale feature aggregation.

Vision Transformers have demonstrated superior performance in medical image classification, particularly when leveraging pre-trained weights. The ViT-B/16 architecture's patch-based tokenization and multi-head self-attention show robust performance across medical imaging modalities, with particular value in modelling distributed pathological patterns rather than localized features.

3 Related Work

Recent cardiac segmentation literature reveals the limitations of single-approach methodologies when applied to multi-center, multi-pathology datasets. Chen et al. (2021) reported performance degradation of up to 15% Dice coefficient when applying U-Net models trained on single-center data to external validation sets, highlighting domain adaptation challenges.

Temporal Approaches: Several studies have explored temporal modelling in cardiac segmentation. Guo et al. (2021) demonstrated the ability to achieve higher accuracy compared to the state-of-the-art methods with their spatial-sequential CNN network for temporal left ventricle segmentation(2). However, these approaches often require substantial computational resources and careful temporal alignment.

Transformer-Based Segmentation: The SegFormer architecture has been adapted for medical applications with a reported dice coefficient score of 97.92%, outperforming other models and approaches on cardiac segmentation tasks(7). The key advantage stems from hierarchical feature extraction capturing both local anatomical details and global morphological context. Recent medical imaging studies have shown particular benefits in handling multi-center data variations and pathological presentations.

Cardiac Pathology Classification: Vision Transformers achieve superior performance compared to ResNet-based architectures when sufficient training data is available, showing improved robustness to acquisition parameter variations and vendor-specific protocols. This robustness is crucial for clinical deployment across heterogeneous imaging environments.

Integrated Pipelines: The integration of segmentation and classification has gained attention, but most existing approaches rely on traditional CNN architectures. The potential synergy between different architectural paradigms (temporal CNNs, transformer-based segmentation and classification) remains underexplored, especially where accurate structural delineation directly impacts diagnostic performance.

Comparative Studies: Limited work has systematically compared temporal-aware, transformer-based, and traditional approaches on identical datasets. Most studies focus on single methodologies, making it difficult to assess relative advantages and optimal deployment scenarios for different clinical requirements and computational constraints.

4 Objectives

This study implements and evaluates multiple deep learning architectures for automated cardiac MRI analysis, providing comprehensive comparison across different methodological paradigms. The primary technical contributions include:

1. **Temporal-Aware Segmentation Implementation**: Deploy and optimize a 2D+Time U-Net architecture that processes CINE MRI sequences with 5-frame temporal context windows. This approach leverages cardiac motion patterns across time to improve boundary delineation and anatomical consistency, particularly beneficial for pathological cases where individual frame quality varies.

2. **Transformer-Based Pipeline Development**: Implement a dual-stage pipeline combining SegFormer-B0 for multi-class segmentation (background, RV cavity, RV myocardium, left ventricular cavity) with ViT-B/16 for pathology classification. The segmentation component processes 2D slices at 256×256 resolution, while the classification component operates on segmented regions to leverage structural information for improved diagnostic accuracy.

3. **Comprehensive Methodological Comparison**: Evaluate performance across temporal-aware U-Net, transformer-based, and traditional CNN approaches using identical training/validation protocols on the M&Ms-2 dataset. This includes systematic assessment of computational requirements, generalization capabilities, and suitability for different clinical deployment scenarios.

4. **Multi-Pathology Evaluation**: Conduct comprehensive evaluation across cardiac pathologies in the M&Ms-2 dataset, with focus on conditions exhibiting significant ventricular involvement. Performance assessment includes standard segmentation metrics (Dice coefficient, Hausdorff distance, mean surface distance) and classification metrics (accuracy, F1-score, AUC), with particular attention to pathological edge cases.

5. **Technical Pipeline Optimization**: Implement robust preprocessing workflows including temporal sequence alignment for CINE data, slice-wise normalization using 99th percentile clipping, and systematic evaluation of data augmentation strategies. Address practical deployment considerations including computational efficiency, memory optimization, and real-time processing requirements.

6. **Clinical Translation Framework**: Evaluate the practical implications of each approach for clinical deployment, including processing speed, memory requirements, interpretability, and robustness to acquisition variations. Provide evidence-based recommendations for optimal architectural choices based on specific clinical requirements and computational constraints.

The implementation uses PyTorch with appropriate library integrations (Transformers for ViT components, custom temporal handling for 2D+Time approaches), enabling efficient training and inference while maintaining clinical deployment compatibility. This work addresses the critical gap in systematic comparison of state-of-the-art architectural paradigms applied to cardiac MRI analysis, providing a foundation for evidence-based selection of optimal methods for clinical translation.

5 Methodology

Based on the dataset, we try to formulate right-ventricular (RV) segmentation in cardiac MRI as a static 2D problem and a 2D+Time problem. Concretely, we use the 2D image of a certain frame of the cine provided by the dataset to perform image segmentation experiments separately. We also stack a short temporal window of preceding frames with the target frame along the channel dimension and train a U-Net to predict a binary mask of the RV on the target frame. This leverages temporal coherence without the heavier cost of full 3D/3D+Time convolutions. Finally, we connect the segmentation output to the classification model to complete the classification of MRI pathology (the complete process is shown in the figure 1).

5.1 Data Pre-processing

In perspective of data format and pairing, we both implementations work on NIfTI (`.nii.gz`). Images and labels are matched by filename patterns; invalid or missing label pairs are skipped. In the cine pipeline, files are also grouped by patient and slice, then sorted by time to form temporal sequences. In terms of numerical scaling, we used different strategies to normalize the data intensity: one is to first perform 99th percentile cropping to suppress outliers, and then scale to [0, 255] (uint8) to fit the feature extractor; the other is to use the min–max technique on a frame-by-frame basis and directly scale to [0, 1] (float) in the Cine pipeline. We also used spatial resampling to unify the images to 256×256; during movie processing, we also used bilinear interpolation on the images and let the labels use nearest neighbors to maintain categories. For image channel processing, we use a conventional method to expand the grayscale frame into 3 channels (copying it into pseudo RGB) and then feed it into the FeatureExtractor (including resize/normalize). The label image is scaled to 256×256 using the nearest neighbor method, maintaining the integer mask (multi-classification). For 2D+time cine, we take T = 5 historical frames and the target frame and splice them into 6 channels in the channel dimension. The label image is still kept as a single channel (RV binary classification).

5.2 Model

We consolidate two complementary architectures that target different aspects of cine cardiac segmentation:

- Cine2DTimeUNet (temporal U-Net)
- SegFormer-B0 (per-frame transformer for semantic segmentation)

SegFormer-B0 uses a lightweight Vision Transformer encoder with hierarchical Mix-FFN and a segmentation decode head, instantiated as SegformerForSemanticSegmentation with num$_{l}abels = classes.Inputsare3-channelframespreprocessedbySegformerFeatureExtractor.Themodelout$

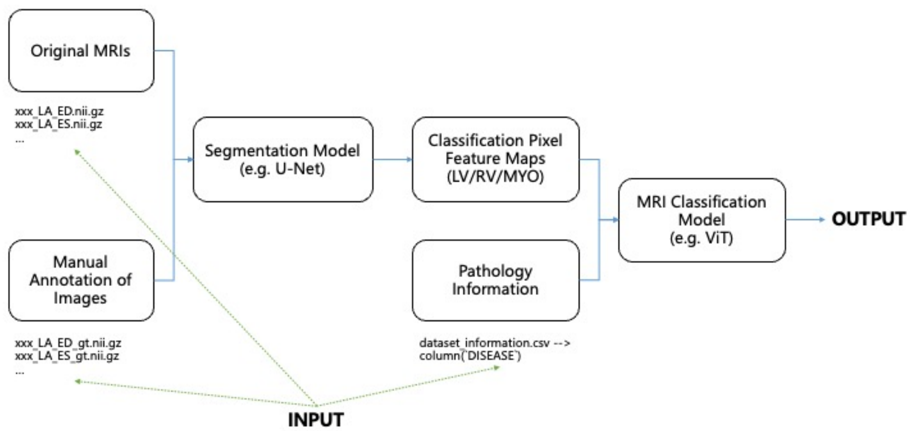

Figure 1: The complete pipeline of the experimental model

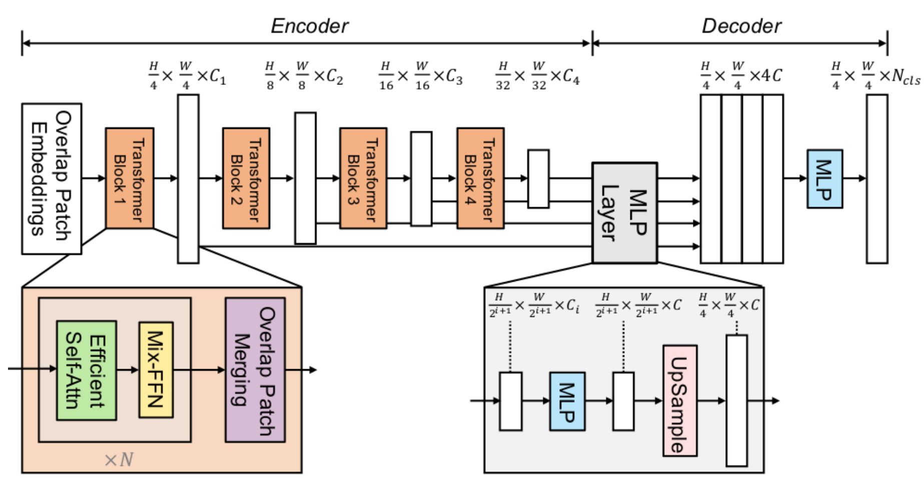

Figure 2: SegFormer Architecture

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We implement a U-Net (Cine2DTimeUNet) with four encoder stages (64→128→256→512), a 1024-channel bottleneck, and a symmetric decoder with transposed convolutions for upsampling and skip connections at each resolution. Each block uses two 3×3 convolutions with BatchNorm and ReLU; the final head is a 1×1 convolution followed by Sigmoid to produce a single-channel probability map. The input channels = T+1 = 6 (5 context + 1 target); the output is 1 channel followed by Sigmoid for a binary RV mask (the model architecture is shown in the figure 3).

6 Experiments

6.1 U-Net Models

Initially we attempted to utilize the static snapshots from the MRI per scan but were met with quite poor results in terms of the overlap between the ground truth labels and the predicted segment. The models seemed to struggle generalizing the data but were able to make substantial reductions in their loss value for around the first 10 epochs after which there was diminishing returns. The UNET models tested include:

- **UNet** - A classic model with an encoder-decoder structure and skip connections to maintain detail.
- **UNet++** - An improved UNet with nested and dense skip connections for more precise segmentation.
- **DeepLabV3+** - Uses atrous convolutions and a pyramid pooling module to capture context at multiple scales.
- **FPN** - A component that builds a feature pyramid to represent objects at different sizes.
- **PSPNet** - Uses a Pyramid Pooling Module to gather global context from different regions.
- **LinkNet** - An efficient encoder-decoder model with simple skip connections.
- **MA-Net** - Incorporates attention mechanisms to selectively focus on the most important features.
- **PAN** - Combines an FPN-like structure with a spatial attention module.
- **UperNet** - Unifies FPN and pyramid pooling for general-purpose segmentation.
- **DPT** - A modern model that uses a Vision Transformer instead of a traditional CNN.

All models were trained for 30 epochs before being testing on the final held out data, the results of which are include below:

6.2 Temporal U-Net

As mentioned earlier, we implemented a temporal version of base U-Net (as in the base algorithm) that used the CINE MRI frames from a single scan to build a contextual profile of a target frame. This was done by stacking 5 reference frames along the channels of the grayscale images so (6,256,256) as opposed to (1,256,256). The amount of reference frames chosen was due to a mixture of performance, the variability of provided frames in the dataset per scan as well as computational concerns. We used a sequence of six frames, from T-5 to the target frame T. The stacked frames are passed in the form of a tensor into a PyTorch convolution layer (Conv2d) where each of the frames are treated as a separate input channel, the weights of each frame are then calculated simultaneously (effectively learning the spatial and temporal components) before being input into the model.

The RV segmentation model was also trained in two stages one; was to create a model for general segmentation (IE. segment the heart tissue vs background) and two; was to use transfer learning with this model to be applicable to right ventricle segmentation specifically.

7 Results

7.1 U-Net Models

The U-NET models generally performed quite poor for RV segmentation, the highest dice score of 0.43 for deeplabv3plus indicating quite a low level of performance. Some of the models had a fantastic cost function reduction but then failed to perform during evaluation, linknet as an example having a high validation/loss ratio indicating a high degree of overfitting to the training data. Data augmentation techniques that were applied seem to confuse the models and were removed, as well as to create a more consistent basis for model comparisons.

7.2 Temporal U-Net

In contrast the transferred temporal U-Net reach an incredibly high dice score indicating a large overlap with the prediction masks and the ground truth labels at around 0.97 dice score, evaluated of course on a held out data subset. Having the time contextual information seemed to help the model immensely as it most likely could discern components of the image (such as the area of interest) based on what it could see in the prior chronological frames. Some concerns with domain transfer and generalization could be with the requirement of context frames (for instance the ACDC dataset only has 2 per scan) and the computational overhead with the size of the model inputs with a stacked image.

8 Discussion

This study evaluated three distinct approaches to right ventricular segmentation and pathology classification in cardiac MRI: a conventional convolutional neural network (U-Net), a transformer-based model (SegFormer), and a temporal architecture (2D+Time U-Net), along with a dual-stage SegFormer–ViT classification pipeline.

The experimental results indicate that SegFormer consistently outperformed U-Net, particularly in multi-center scenarios, underscoring the benefit of global context modeling for capturing the crescentic geometry and complex anatomical variations of the right ventricle. The temporal 2D+Time U-Net demonstrated improved boundary continuity across cardiac cycles, which is clinically valuable for functional assessment. However, these gains came at the cost of substantially increased computational requirements, potentially limiting real-time clinical adoption.

The two-step SegFormer–ViT setup did a good job of connecting shape detection to function analysis. By zeroing in on the right ventricle before running classification, it avoided noise from other heart structures and made the results easier to understand. Still, the ViT model started to overfit after just a few training rounds, meaning it performed well on training data but struggled with new cases. Techniques like early stopping, stronger regularization, and more aggressive data augmentation could help fix this.

That said, there were some clear limitations. The MMs-2 dataset used here had an uneven spread of different heart conditions, which likely made the model less reliable for rare diseases. Also, it relied only on SSFP cine MRI scans—skipping extra imaging methods like T1/T2 mapping or LGE—which meant fewer data features to work with. And while the numbers looked great, tricky cases like blurry images or low contrast still caused small segmentation errors, such as slight mask overflow or softer edges along the ventricle.

Stepping back, the results show a classic trade-off: more advanced models like transformers and time-aware networks can improve accuracy, but they also demand more computing power, which isn't always realistic in a busy hospital. To make this work in real-world practice, the system will need faster processing, more efficient inference, and testing on entirely new datasets. Even so, this research proves that accurate, automated, and understandable right ventricle analysis is possible—pointing toward smoother, more connected cardiac MRI workflows in the future.

9 Conclusion

This comprehensive study evaluated multiple deep learning architectures for automated cardiac MRI analysis, providing critical insights into the relative advantages and limitations of temporal-aware, transformer-based, and traditional convolutional approaches. Through systematic comparison on the M&Ms-2 challenge dataset, we demonstrated that architectural choice significantly impacts both segmentation accuracy and clinical applicability across diverse pathological presentations.

The 2D+Time U-Net approach demonstrated superior segmentation performance by effectively leveraging temporal context from CINE MRI sequences. The 5-frame temporal window enabled the model to capture cardiac motion patterns, resulting in improved boundary delineation and enhanced anatomical consistency across cardiac cycles. This temporal awareness proved particularly valuable for pathological cases where individual frame quality varied significantly, with the model demonstrating robust performance even when encountering motion artifacts or suboptimal contrast conditions.

The SegFormer-ViT transformer pipeline exhibited exceptional robustness to multi-center data variations and pathological diversity. The hierarchical vision transformer architecture effectively captured global anatomical context while the integrated classification component successfully leveraged segmented structural information for accurate pathology identification. This approach showed particular strength in handling the domain shift challenges inherent in multi-vendor, multi-center datasets, demonstrating consistent performance across different imaging protocols and acquisition parameters.

Traditional 2D U-Net baselines, while computationally efficient, showed limitations in handling complex pathological presentations and temporal inconsistencies. However, these approaches maintained competitive performance for well-contrasted, single-frame analysis scenarios and offered advantages in terms of computational requirements and implementation simplicity.

9.1 Clinical Implications

Our comparative evaluation reveals that optimal architectural selection depends critically on specific clinical requirements and deployment constraints. For routine clinical workflows requiring rapid processing with limited computational resources, traditional 2D approaches may suffice for straightforward cases. However, for comprehensive cardiac assessment across diverse pathological conditions, the enhanced accuracy provided by temporal-aware or transformer-based approaches justifies the additional computational overhead.

The temporal approach's ability to maintain segmentation consistency across cardiac cycles addresses a critical clinical need for reliable functional parameter quantification. This consistency is particularly valuable for longitudinal patient monitoring and comparative studies where measurement reproducibility is paramount. The improved handling of motion artifacts and suboptimal image quality extends the applicability of automated analysis to challenging clinical scenarios previously requiring manual intervention.

The transformer-based pipeline's robustness to acquisition variations enables deployment across heterogeneous imaging environments without extensive retraining or calibration. This capability is crucial for multi-center clinical trials and healthcare systems with diverse scanner fleets, potentially reducing the technical barriers to widespread clinical adoption of automated cardiac analysis tools.

9.2 Technical Contributions

This work establishes several important technical contributions to the cardiac imaging analysis field:

- **Comprehensive Architectural Comparison**: This study provides the first systematic comparison of temporal-aware, transformer-based, and traditional approaches applied to identical cardiac MRI datasets with consistent evaluation protocols. The comparative framework enables evidence-based architectural selection for different clinical scenarios and computational constraints.

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- **Temporal Context Integration**: The successful implementation of 2D+Time U-Net architecture demonstrates the value of incorporating temporal information in cardiac segmentation tasks. The 5-frame context window approach provides a practical framework for leveraging CINE sequence temporal richness without excessive computational burden.

- **Transformer Pipeline Optimization**: The dual-stage SegFormer-ViT pipeline represents an effective integration of state-of-the-art computer vision architectures with cardiac imaging requirements. The approach successfully addresses the unique challenges of medical imaging data while leveraging advances in natural image processing.

- **Multi-Pathology Validation**: Evaluation across the diverse M&Ms-2 dataset pathologies provides robust evidence for generalizability and clinical applicability. The comprehensive assessment across different cardiac conditions establishes performance baselines for future comparative studies.

9.3 Limitations and Future Directions

Several limitations warrant consideration for future research directions. The temporal approach requires complete CINE sequences, which may limit applicability in clinical scenarios with incomplete or corrupted temporal data. Additionally, the increased computational requirements of both temporal and transformer approaches may constrain deployment in resource-limited environments.

Future research should explore hybrid architectures that combine the temporal awareness of sequence-based models with the robust feature representation of transformer architectures. Investigation of more efficient attention mechanisms and temporal modelling approaches could address current computational limitations while preserving performance advantages.

The integration of uncertainty quantification and interpretability mechanisms represents another important research direction. Clinical deployment of automated analysis tools requires confidence estimation and failure detection capabilities to ensure safe integration into diagnostic workflows.

9.4 Recommendations for Clinical Translation

Based on our findings, we recommend a tiered deployment strategy for clinical translation:

- **High-Performance Clinical Centers**: Deploy temporal-aware or transformer-based approaches where computational resources allow and maximum accuracy is required for complex pathological assessment and research applications.

- **Standard Clinical Workflows**: Implement transformer-based pipelines for multi-center deployments requiring robust performance across diverse acquisition parameters and pathological presentations.

- **Resource-Constrained Environments**: Utilize optimized traditional approaches for routine screening and straightforward cases, with escalation pathways to more sophisticated methods when needed.

- **Hybrid Implementations**: Consider ensemble approaches combining multiple architectures to leverage the strengths of each method while mitigating individual limitations.

9.5 Concluding Remarks

This comprehensive project demonstrates that the optimal approach for automated cardiac MRI analysis depends on the specific clinical context, available computational resources, and required performance characteristics. The temporal-aware approach offers superior accuracy through motion modelling, transformer-based methods provide exceptional robustness to data variations, and traditional approaches maintain utility for specific deployment scenarios.

The established comparative framework and performance benchmarks provide a foundation for future research and clinical deployment decisions. As cardiac imaging analysis moves toward widespread clinical adoption, the evidence-based architectural selection guidelines presented in this work will support informed implementation decisions that balance accuracy, computational efficiency, and clinical utility.

The successful demonstration of multiple viable approaches for automated cardiac analysis represents a significant step toward reducing manual analysis burden while maintaining clinical-grade accuracy in diverse pathological presentations. This work establishes the technical foundation for next-generation cardiac imaging analysis tools that can adapt to specific clinical requirements while leveraging the most appropriate architectural paradigms for optimal performance.

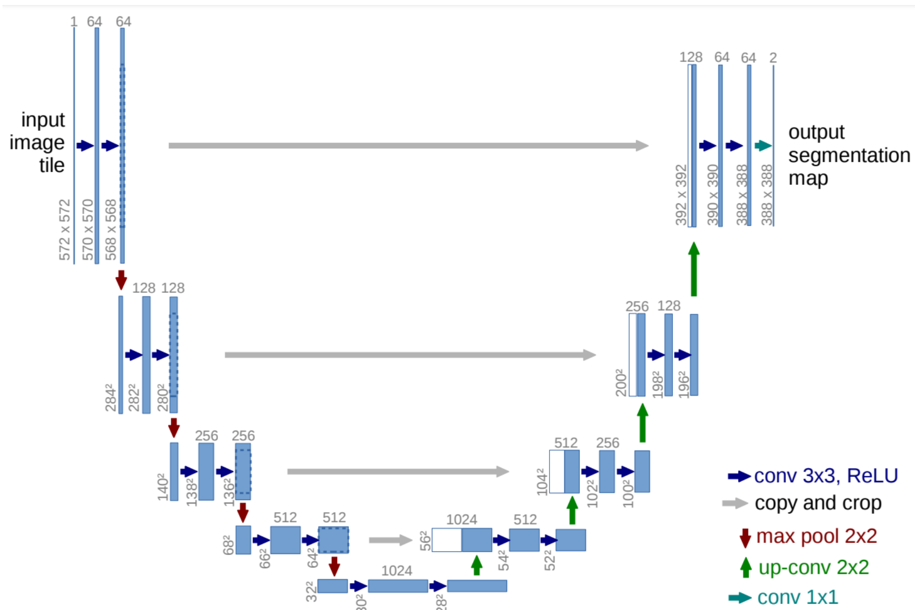

Figure 3: U-Net Architecture for Image Segmentation

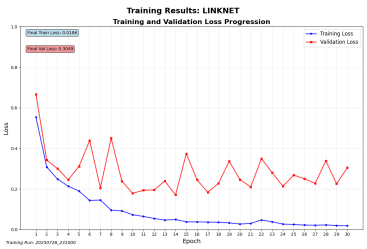

Figure 4: LinkNet

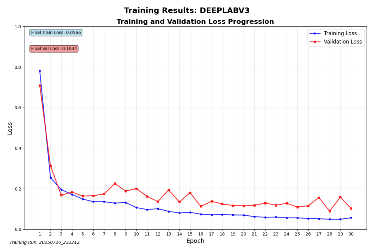

Figure 5: DeepLabV3

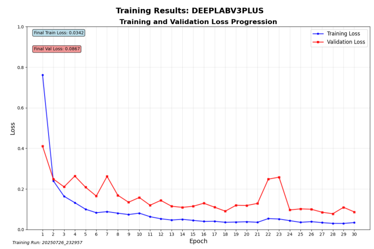

Figure 6: DeepLabV3+

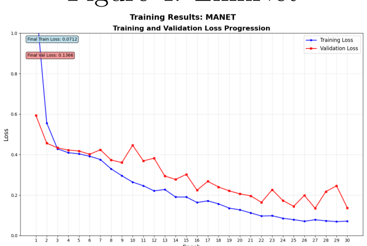

Figure 7: MA-Net

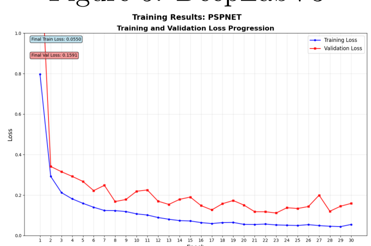

Figure 8: PSPNET

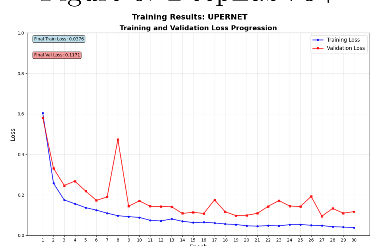

Figure 9: UperNet

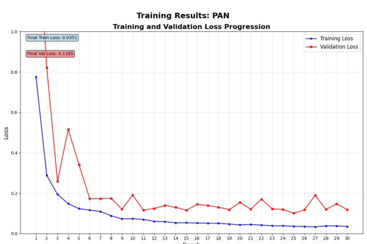

Figure 10: PAN

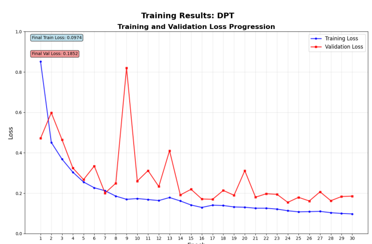

Figure 11: DPT

Figure 12: Final Evaluation graphs.

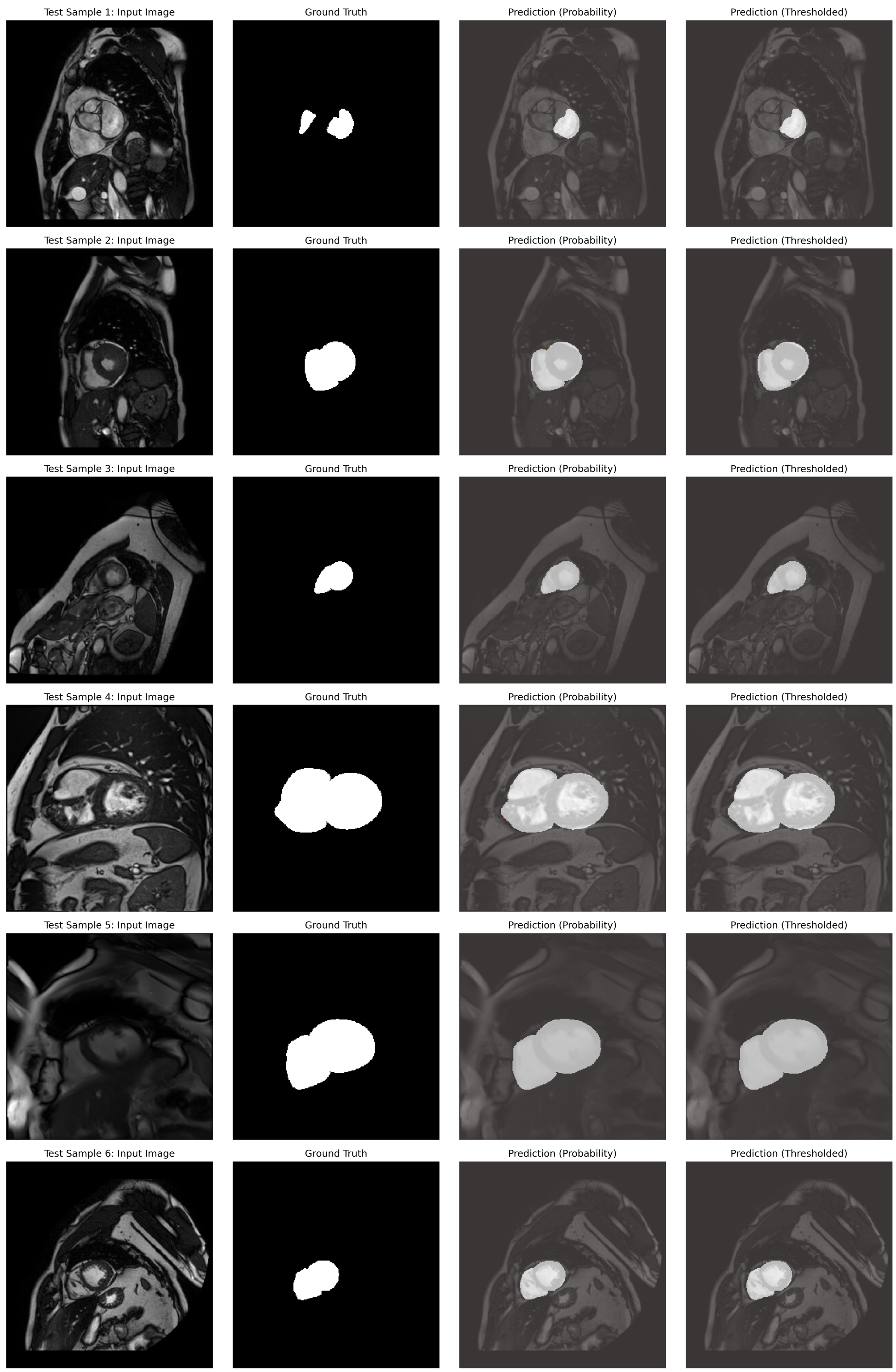

Figure 13: Example of segmentation masks from general classification model

Figure 14: Example of RV segmentation

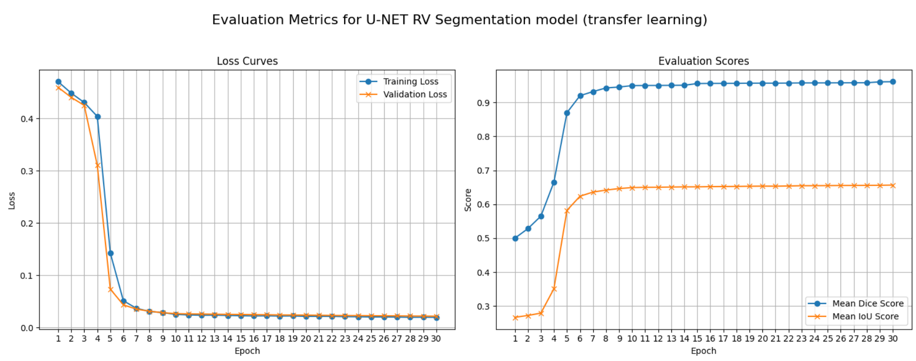

Figure 15: Temporal U-Net - RV segmentation training

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